# MATH-3IA3 Assignment 2

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## 2.5.6.) If $I_1 \supseteq I_2 \supseteq \ldots \supseteq I_n \supseteq \ldots$ is a nested sequence of intervals and if $I_n = [a_n, b_n]$, show that $a_1 \leq a_2 \leq \ldots \leq a_n \leq \ldots$ and $b_1 \geq b_2 \geq \ldots \geq b_n \geq \ldots$

We prove both statements by induction. Let

$$P_1(n) : a_1 \leq a_2 \leq \ldots \leq a_{n+1}, \quad \text{and} \quad P_2(n) : b_1 \geq b_2 \geq \ldots \geq b_{n+1}.$$

### a.) Prove $P_1(n)$

I.) Base case (Prove $P_1(1)$ is true)

$P_1(1)$ boils down to the statement $a_1 \leq a_2$. For the purposes of contradiction, let's assume $P_1(1)$ is not true. We know

$$a_1 = \inf(I_1) \quad \text{and} \quad a_2 = \inf(I_2)$$

by definition of intervals, and since both intervals are closed,

$$a_1 \in I_1 \quad \text{and} \quad a_2 \in I_2.$$

Since $I_1 \supseteq I_2$, we know that $a_2 \in I_1$. However, if $a_1 > a_2$, then $a_1$ cannot be the infimum of $I_1$ since there exists an element smaller than it in the set. Therefore, by contradiction, it must be that

$$a_1 \leq a_2,$$

and thus $P_1(1)$ is true.

II.) Induction Hypothesis (State $P_1(k)$)

$$P_1(k) : a_1 \leq a_2 \leq \ldots \leq a_{k+1}.$$

III.) Prove $P_1(k) \implies P_1(k+1)$

We need to show that

$$P_1(k+1) : a_1 \leq a_2 \leq \ldots \leq a_{k+2}.$$

Assuming $P_1(k)$, we know the statement

$$a_1 \leq a_2 \leq \ldots \leq a_{k+1}$$

is true. Now once again appealing to our earlier argument, we know that if

$$I_{k+1} \supseteq I_{k+2},$$

then

$$a_{k+2} \in I_{k+1} = [a_{k+1}, b_{k+1}],$$

and hence

$$a_{k+1} \leq a_{k+2}.$$

Combining this with our induction hypothesis, we have

$$a_1 \leq a_2 \leq \ldots \leq a_{k+1} \leq a_{k+2}.$$

Therefore, $P_1(k+1)$ holds true. By the principle of mathematical induction, $P_1(n)$ is true for all $n$.

## b.) Prove $P_2(n)$

I.) Base case (Prove $P_2(1)$ is true)

$P_2(1)$ boils down to the statement $b_1 \geq b_2$. Since $I_1 \supseteq I_2$, we know that

$$b_2 \in I_1 = [a_1, b_1].$$

Hence

$$a_1 \leq b_2 \leq b_1,$$

and therefore

$$b_1 \geq b_2.$$

Thus $P_2(1)$ is true.

II.) Induction Hypothesis (State $P_2(k)$)

$$P_2(k) : b_1 \geq b_2 \geq \ldots \geq b_{k+1}.$$

III.) Prove $P_2(k) \implies P_2(k+1)$

We need to show that

$$P_2(k+1) : b_1 \geq b_2 \geq \ldots \geq b_{k+2}.$$

Assuming $P_2(k)$, we know that

$$b_1 \geq b_2 \geq \ldots \geq b_{k+1}.$$

Now since

$$I_{k+1} \supseteq I_{k+2},$$

it follows that

$$b_{k+2} \in I_{k+1} = [a_{k+1}, b_{k+1}],$$

and so

$$a_{k+1} \leq b_{k+2} \leq b_{k+1}.$$

This implies

$$b_{k+1} \geq b_{k+2}.$$

Combining this with our induction hypothesis, we have

$$b_1 \geq b_2 \geq \ldots \geq b_{k+1} \geq b_{k+2}.$$

Thus, $P_2(k+1)$ holds true. By the principle of mathematical induction, $P_2(n)$ is true for all $n$.

## 3.1.17.) Show that $\lim_{n\to\infty} \frac{2^n}{n!} = 0$. Hint: If $n \geq 3$, then $0 < \frac{2^n}{n!} \leq 2(\frac{2}{3})^{n-2}$.

*Proof.* Let $\epsilon > 0$ be arbitrary. To prove that $\lim_{n\to\infty} \frac{2^n}{n!} = 0$, we must show that there exists some $N \in \mathbb{N}$ such that $n \geq N \implies |\frac{2^n}{n!} - 0| < \epsilon$. To start, we begin by setting $N_1 = 3$. Thus, using the hint given in the question, for all $n \geq N_1$, we have:

$$\begin{aligned}
|\frac{2^n}{n!} - 0| &= |\frac{2^n}{n!}| \\
&\leq |2(\frac{2}{3})^{n-2}| \\
&= |\frac{9}{2}(\frac{2}{3})^n| \\
&= \frac{9}{2}(\frac{2}{3})^n
\end{aligned}$$

Now let's consider the sequence $(\frac{2}{3})^n$. Our goal is to prove that this sequence too converges to zero, because if it does, we can make that term arbitrary small in our previous expression and our proof that $\lim_{n\to\infty} \frac{2^n}{n!} = 0$ is complete. Since $0 < \frac{2}{3} < 1$, we know that for all $x, y \in \mathbb{R}, \text{with} x < y, (\frac{2}{3})^y \leq (\frac{2}{3})^x$. Since this fact obviously extends to the natural numbers as well, we know the sequence is monotonously decreasing, and clearly upper and lower bounded by 1 and 0 respectively. Thus by the monotone convergence theorem, we know that the sequence $\frac{2}{3}^n$ must converge. Using this, I conjecture and will proceed to prove that $\lim_{n\to\infty} \frac{2^n}{n!} = 0$. Letting $N_2 > \log_{\frac{2}{3}} \frac{2}{9}\epsilon$, we know that for all $n \geq N_2$:

$$\begin{aligned}
|(\frac{2}{3})^n - 0| &= |(\frac{2}{3})^n| \\
&\leq |(\frac{2}{3})^{N_2}| \\
&< |(\frac{2}{3}^{\log_{\frac{2}{3}}(\frac{2}{9}\epsilon)}) \\
&= |\frac{2}{9}\epsilon| \\
&= \frac{2}{9}\epsilon
\end{aligned}$$

Therefore, by definition of convergence, we know that $\lim_{n\to\infty} (\frac{2}{3})^n = 0$, furthermore by letting $N = max\{N_1, N_2\}$, for all $n \geq N$, we have:

$$\begin{aligned}
|\frac{2^n}{n!} - 0| &= |\frac{2^n}{n!}| \\
&\leq |2(\frac{2}{3})^{n-2}| \\
&= |\frac{9}{2}(\frac{2}{3})^n| \\
&= \frac{9}{2}(\frac{2}{3})^n \\
&< \frac{9}{2}(\frac{2}{9}\epsilon) \\
&= \epsilon
\end{aligned}$$

Hence, we have successfully shown that $\lim_{n\to\infty} \frac{2^n}{n!} = 0$. □

## 3.2.17.)

a.) Give an example of a convergent sequence $(x_n)$ of positive numbers with $\lim_{n\to\infty}\frac{(x_{n+1})}{(x_n)} = 1$.

The constant sequence $(x_n) = 1$ is trivially convergent to 1 and has the property that $\lim_{n\to\infty}\frac{(x_{n+1})}{(x_n)} = 1$ since:

$$\begin{aligned}\lim_{n\to\infty}\frac{(x_{n+1})}{(x_n)} &= \lim_{n\to\infty}\frac{1}{1}\\ &= 1\end{aligned}$$

b.) Give an example of a divergent sequence with this property. (Thus, this property cannot be used as a test for convergence.)

The sequence $(x_n) = n$ is an example of a clearly divergent sequence (since it is clearly not bounded) with the property that $\lim_{n\to\infty}\frac{(x_{n+1})}{(x_n)} = 1$ since (by use of algebraic limit rules):

$$\begin{aligned}\lim_{n\to\infty}\frac{(x_{n+1})}{(x_n)} &= \lim_{n\to\infty}\frac{n+1}{n}\\ &= \lim_{n\to\infty}(1+\frac{1}{n})\\ &= \lim_{n\to\infty}1 + \lim_{n\to\infty}\frac{1}{n}\\ &= 1+0\\ &= 1\end{aligned}$$

## 3.2.18.) Let $X = (x_n)$ be a sequence of positive real numbers such that $\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = L > 1$. Show that X is not a bounded sequence and hence is not convergent.

*Proof.* Assume, for the purposes of contraction, that X is a bounded sequence. Then, by definition of boundedness, we know there must exist some $M > 0$ such that $\forall n \in \mathbb{N}, |x_n| < M$. Furthermore, by definition of convergence, for arbitrary $\epsilon > 0$ we know that there exists an $N \in \mathbb{N}$ such that $n \geq N \implies$

$$|\frac{x_{n+1}}{x_n} - L| < \frac{\epsilon}{M}$$

However, that would lead to the following implications:

$$\begin{aligned}
|\frac{x_{n+1}}{x_n} - L| < \frac{\epsilon}{M} \implies \\
|\frac{x_{n+1} - Lx_n}{x_n}| < \frac{\epsilon}{M} \implies \\
|x_{n+1} - Lx_n| < \frac{\epsilon}{M}|x_n| \implies \\
|x_{n+1} - Lx_n| < \frac{\epsilon}{M}M \implies \\
|x_{n+1} - Lx_n| < \epsilon
\end{aligned}$$

Thus, we know that for all $n \geq N, x_{n+1} = Lx_n$. Therefore, we can arrive at the conclusion that for all such n, $x_{n+k} = L^k x_n$ by recursion. Consider the case where $n = N$. Since $L > 1$, we know that there exists some $K \in \mathbb{N}$ such that $n \geq K \implies L^n \geq \frac{M}{x_N}$. However, this would mean, by our earlier property that:

$$\begin{aligned}
x_{N+K} &= L^K x_N \\
&\geq \frac{M}{x_N} x_N \\
&= M
\end{aligned}$$

However, we had previously assumed that our sequence was bounded by M and thus $\forall n \in \mathbb{N}\ |x_n| < M$. Thus, by contradiction, our sequence X cannot be bounded and therefore, cannot be convergent. Thus, X is divergent. $\square$

## 3.3.12.) Establish the convergence and find the limits of the following sequences.

### a.) $((1 + \frac{1}{n})^{n+1})$

Let $(x_n) = 1$, $(y_n) = \frac{1}{n}$, and $z_n = (1 + \frac{1}{n})^n$. We can therefore express $((1 + \frac{1}{n})^{n+1})$ as:

$$\begin{aligned} ((1 + \frac{1}{n})^{n+1}) &= (1 + \frac{1}{n})(1 + \frac{1}{n})^n \\ &= (x_n + y_n)(z_n) \\ &= (x_n)(z_n) + (y_n)(z_n) \end{aligned}$$

Thus, we can express $((1 + \frac{1}{n})^{n+1})$ as the sum of two sequences being multiplied together. However, we know that each such limit exists. $\lim_{n\to\infty} x_n = \lim_{n\to\infty} 1$ is trivially 1, $\lim_{n\to\infty} y_n = \lim_{n\to\infty} \frac{1}{n} = 0$ which was proven in example 3.1.6, and $\lim_{n\to\infty} z_n = \lim_{n\to\infty} (1 + \frac{1}{n})^n = e_n$, as defined in example 3.3.6. Thus, by theorem 3.2.3, we know that $\lim_{n\to\infty}((x_n)(z_n))$ exists and $\lim_{n\to\infty}((y_n)(z_n))$ exists. So, once again appealing to theorem 3.2.3, we know $\lim_{n\to\infty}((x_n)(z_n) + (y_n)(z_n))$ also exists, and since $((1 + \frac{1}{n})^{n+1}) = (x_n)(z_n) + (y_n)(z_n)$, we therefore know $\lim_{n\to\infty}((1 + \frac{1}{n})^{n+1})$ exists. Appealing to theorem 3.2.3 one more time, we know that:

$$\begin{aligned} \lim_{n\to\infty}((1 + \frac{1}{n})^{n+1}) &= \lim_{n\to\infty}((x_n)(z_n) + (y_n)(z_n)) \\ &= \lim_{n\to\infty} x_n \lim_{n\to\infty} z_n + \lim_{n\to\infty} y_n \lim_{n\to\infty} z_n \\ &= (1)(e_n) + (0)(e_n) \\ &= e_n \end{aligned}$$

### b.) $((1 + \frac{1}{n})^{2n})$

Consider that:

$$\begin{aligned} (1 + \frac{1}{n})^{2n} &= ((1 + \frac{1}{n})^n)^2 \\ &= (1 + \frac{1}{n})^n(1 + \frac{1}{n})^n \end{aligned}$$

By example 3.3.6, we know that $\lim_{n\to\infty}(1 + \frac{1}{n})^n$ exists and equals $e_n$. Therefore, by theorem 3.2.3, we know that $((1 + \frac{1}{n})^{2n})$ exists and that:

$$\begin{aligned} \lim_{n\to\infty}((1 + \frac{1}{n})^{2n}) &= \lim_{n\to\infty}((1 + \frac{1}{n})^n(1 + \frac{1}{n})^n) \\ &= \lim_{n\to\infty}(1 + \frac{1}{n})^n \lim_{n\to\infty}(1 + \frac{1}{n})^n \\ &= (e_n)(e_n) \\ &= (e_n)^2 \end{aligned}$$

### c.) $((1 + \frac{1}{n+1})^n)$

Here we will prove that the sequence converges by squeeze theorem. Consider the sequences $(x_n) = e_n$ and $(y_n) = (1 + \frac{1}{n})^n$. Clearly, $\lim_{n\to\infty}(x_n) = e_n$ and by example 3.3.6, we know $\lim_{n\to\infty}(y_n) = e_n$. We also know from example 3.3.6 that $(y_n)$ is a decreasing sequence, and therefore that $e_n \leq (1 + \frac{1}{n})^n \quad \forall n \in \mathbb{N}$ by theorem 3.3.2.b, and because $(n+1) \in \mathbb{N}$, we therefore know $e_n = (x_n) \leq (1 + \frac{1}{n+1})^n$. Now, we know that $\frac{1}{n+1} \leq \frac{1}{n}$, and therefore that $1 + \frac{1}{n+1} \leq 1 + \frac{1}{n}$ and since $n > 0$, we know that $(1 + \frac{1}{n+1})^n \leq (1 + \frac{1}{n})^n$.

Therefore, since $(x_n) \leq (1 + \frac{1}{n+1})^n \leq (y_n)$, and $(x_n) = (y_n) = e_n$ we can appeal to squeeze theorem to state both that $\lim_{n\to\infty}(1 + \frac{1}{n+1})^n$ exists and equals $e_n$.

## d.) $(1 - \frac{1}{n})^n$

We will prove the convergence of $(1 - \frac{1}{n})^n$ by the monotone convergence theorem.

### I.) Prove that $(1 - \frac{1}{n})^n$ is bounded

Since $1 - \frac{1}{n} \leq 1 \quad \forall n \in \mathbb{N}$, we know $(1 - \frac{1}{n})^n \leq (1)^n = 1$. Furthermore, since $0 \leq 1 - \frac{1}{n} \quad \forall n \in \mathbb{N}$, then $0 = 0^n \leq (1 - \frac{1}{n})^n$.

Therefore, since $0 \leq (1 - \frac{1}{n})^n \leq 1$, $(1 - \frac{1}{n})^n$ is bounded.

### II.) Prove that $(1 - \frac{1}{n})^n$ is monotone

Using the property that $\frac{1}{n} > \frac{1}{n+1}$ we can derive that $1 - \frac{1}{n} < 1 - \frac{1}{n+1}$. Also, since $0 \leq 1 - \frac{1}{n} \leq 1 \quad \forall n \in \mathbb{N}$, we know that $(1 - \frac{1}{n})^{n+1} \leq (1 - \frac{1}{n})^n$. Therefore, we know:

$$\begin{aligned}
(1 - \frac{1}{n+1})^n + 1 &= (1 - \frac{1}{n+1})^{n+1} \\
&\geq (1 - \frac{1}{n})^{n+1} \\
&\geq (1 - \frac{1}{n})^n
\end{aligned}$$

Therefore, $(1 - \frac{1}{n})^n$ is monotone.

Thus, since $(1 - \frac{1}{n})^n$ is both bounded and monotone, by monotone convergence theorem, the sequence $(1 - \frac{1}{n})^n$ converges.

We have already established that $(1 - \frac{1}{n})^n$ converges by the Monotone Convergence Theorem. To determine what it converges to, we will relate it to the sequence $(1 + \frac{1}{n})^n$, which we know converges to $e_n$.

Observe that:

$$\begin{aligned}
(1 - \tfrac{1}{n})^n &= \frac{1}{(1 - \frac{1}{n})^{-n}} \\
&= \frac{1}{\left(\frac{1}{1 - \frac{1}{n}}\right)^n}.
\end{aligned}$$

Now rewrite the fraction in the denominator:

$$\begin{aligned}
\frac{1}{1 - \frac{1}{n}} &= \frac{n}{n-1} \\
&= 1 + \frac{1}{n-1}.
\end{aligned}$$

Hence,

$$(1 - \tfrac{1}{n})^n = \frac{1}{\left(1 + \frac{1}{n-1}\right)^n}.$$

We already know from Example 3.3.6 that:

$$\lim_{n\to\infty} \left(1 + \tfrac{1}{n-1}\right)^{n-1} = e_n.$$

We can now express our term in a way that uses this known limit:

$$(1-\tfrac{1}{n})^n = \frac{1}{\left[\left(1+\frac{1}{n-1}\right)^{n-1}\right]^{\frac{n}{n-1}}}.$$

Since $\frac{n}{n-1} \to 1$ as $n \to \infty$, and since $\left(1+\frac{1}{n-1}\right)^{n-1} \to e_n$, we can combine these two facts to conclude:

$$\lim_{n\to\infty} (1-\tfrac{1}{n})^n = \frac{1}{e_n}.$$

Therefore, $(1-\frac{1}{n})^n$ converges to $\frac{1}{e_n}$